

Matter around us

MCQ

1. Matter is anything that has

- a) Colour and shape
- b) Mass and occupies space
- c) Energy only
- d) Feelings

Answer: b) Mass and occupies space

2. Which of the following is NOT matter?

- a) Air
- b) Water
- c) Love
- d) Sand

Answer: c) Love

3. Matter is made up of tiny particles called

- a) Cells
- b) Molecules
- c) Atoms
- d) Ions

Answer: c) Atoms

4. An atom is best defined as

- a) A visible particle
- b) The smallest unit of matter
- c) A molecule of matter
- d) A compound

Answer: b) The smallest unit of matter

5. Which of the following is an example of matter?

- a) Happiness
- b) Light
- c) Water
- d) Sound

Answer: c) Water

6. One drop of water contains approximately

- a) 10^5 particles
- b) 10^8 particles
- c) 10^{21} particles
- d) 10^3 particles

Answer: c) 10^{21} particles

7. A molecule is formed when atoms are

- a) Mixed physically
- b) Loosely placed
- c) Chemically bonded
- d) Melted

Answer: c) Chemically bonded

8. All compounds are molecules but

- a) All molecules are elements
- b) All molecules are compounds
- c) All atoms are molecules
- d) All molecules are not compounds

Answer: d) All molecules are not compounds

9. Atomicity refers to

- a) Number of molecules

- b) Number of electrons
- c) Number of atoms in a molecule
- d) Atomic number

Answer: c) Number of atoms in a molecule

10. Which is a monoatomic molecule?

- a) Oxygen
- b) Hydrogen
- c) Helium
- d) Water

Answer: c) Helium

11. A diatomic molecule contains

- a) One atom
- b) Two atoms
- c) Three atoms
- d) Many atoms

Answer: b) Two atoms

12. Which of the following is a diatomic molecule?

- a) CO_2
- b) H_2
- c) CH_4
- d) H_2SO_4

Answer: b) H_2

13. H_2O is an example of

- a) Diatomic molecule
- b) Monoatomic molecule
- c) Triatomic molecule

d) Polyatomic molecule

Answer: c) Triatomic molecule

14. Which of the following is a polyatomic molecule?

a) O_2

b) N_2

c) CH_4

d) He

Answer: c) CH_4

15. A homoatomic molecule contains

a) Different kinds of atoms

b) Only ions

c) Same kind of atoms

d) Only metals

Answer: c) Same kind of atoms

16. Which is a heteroatomic molecule?

a) O_3

b) S_8

c) P_4

d) CO_2

Answer: d) CO_2

17. Matter is broadly classified into

a) Solids and liquids

b) Atoms and molecules

c) Pure substances and mixtures

d) Metals and non-metals

Answer: c) Pure substances and mixtures

18. A pure substance has

- a) Variable composition
- b) Fixed composition
- c) Many particles
- d) No particles

Answer: b) Fixed composition

19. Elements and compounds are types of

- a) Mixtures
- b) Pure substances
- c) Solutions
- d) Suspensions

Answer: b) Pure substances

20. An element cannot be broken down by

- a) Physical methods
- b) Chemical reactions
- c) Heating
- d) Cooling

Answer: b) Chemical reactions

21. All atoms of an element have the same

- a) Size only
- b) Mass only
- c) Atomic number
- d) Shape

Answer: c) Atomic number

22. Total number of known elements is

- a) 92

b) 105

c) 118

d) 120

Answer: c) 118

23. Elements are classified into metals, non-metals and

a) Compounds

b) Mixtures

c) Metalloids

d) Molecules

Answer: c) Metalloids

24. Metals are generally

a) Brittle

b) Dull

c) Good conductors

d) Poor conductors

Answer: c) Good conductors

25. The property of metals to be beaten into sheets is

a) Ductility

b) Malleability

c) Hardness

d) Brittleness

Answer: b) Malleability

26. Which metal is used in thermometers?

a) Iron

b) Aluminium

c) Mercury

d) Copper

Answer: c) Mercury

27. Which of the following is NOT a property of metals?

a) Lustrous

b) Sonorous

c) Poor conductor

d) Ductile

Answer: c) Poor conductor

28. Non-metals are generally

a) Lustrous

b) Sonorous

c) Brittle

d) Ductile

Answer: c) Brittle

29. Which non-metal is a good conductor of electricity?

a) Sulphur

b) Carbon (graphite)

c) Phosphorus

d) Iodine

Answer: b) Carbon (graphite)

30. Which non-metal is liquid at room temperature?

a) Chlorine

b) Bromine

c) Iodine

d) Carbon

Answer: b) Bromine

31. Oxygen is mainly used for

- a) Fertilizers
- b) Breathing
- c) Jewellery
- d) Painting

Answer: b) Breathing

32. Elements showing properties of both metals and non-metals are called

- a) Metals
- b) Non-metals
- c) Metalloids
- d) Compounds

Answer: c) Metalloids

33. Silicon is an example of

- a) Metal
- b) Non-metal
- c) Metalloid
- d) Compound

Answer: c) Metalloid

34. A compound is formed by

- a) Physical mixing
- b) Chemical bonding
- c) Mechanical process
- d) Heating only

Answer: b) Chemical bonding

35. Elements in a compound are present in

- a) Variable ratio

- b) Any ratio
- c) Fixed ratio
- d) Random ratio

Answer: c) Fixed ratio

36. Water is a compound because

- a) It contains atoms
- b) It has fixed composition
- c) It is liquid
- d) It evaporates

Answer: b) It has fixed composition

37. Properties of a compound are

- a) Same as elements
- b) Average of elements
- c) Different from elements
- d) Always metallic

Answer: c) Different from elements

38. Constituents of a compound can be separated by

- a) Filtration
- b) Evaporation
- c) Chemical methods
- d) Hand picking

Answer: c) Chemical methods

39. An element symbol represents

- a) One molecule
- b) One compound
- c) One atom

d) One mixture

Answer: c) One atom

40. Element symbols are standardized by

a) WHO

b) UNESCO

c) IUPAC

d) ISRO

Answer: c) IUPAC

41. The symbol Na comes from the

a) English name

b) Greek name

c) Latin name

d) German name

Answer: c) Latin name

42. A chemical formula shows

a) Shape of molecule

b) Colour of substance

c) Composition of molecule

d) Size of atom

Answer: c) Composition of molecule

43. The formula of magnesium chloride is

a) MgCl

b) MgCl_2

c) Mg_2Cl

d) Mg_2Cl_2

Answer: b) MgCl_2

44. Atomicity of water is

- a) 1
- b) 2
- c) 3
- d) 4

Answer: c) 3

45. Atomicity of sulphuric acid (H_2SO_4) is

- a) 6
- b) 7
- c) 8
- d) 9

Answer: b) 7

46. The main elements in the human body are

- a) Iron and copper
- b) Oxygen and nitrogen
- c) Oxygen, carbon, hydrogen, nitrogen
- d) Gold and silver

Answer: c) Oxygen, carbon, hydrogen, nitrogen

47. Air is mainly composed of

- a) Oxygen and carbon dioxide
- b) Nitrogen and oxygen
- c) Hydrogen and oxygen
- d) Nitrogen and argon

Answer: b) Nitrogen and oxygen

48. Trace gases in air include

- a) Helium and neon

- b) Iron and zinc
- c) Mercury and bromine
- d) Sodium and potassium

Answer: a) Helium and neon

49. A state of matter describes

- a) Shape only
- b) Behaviour of particles
- c) Colour of substance
- d) Chemical nature

Answer: b) Behaviour of particles

50. The three common states of matter are

- a) Solid, gas, plasma
- b) Solid, liquid, gas
- c) Liquid, gas, plasma
- d) Solid, liquid, vapour

Answer: b) Solid, liquid, gas

51. Solids have

- a) Definite shape and volume
- b) No definite shape
- c) No definite volume
- d) Very weak forces

Answer: a) Definite shape and volume

52. Liquids have

- a) Definite shape only
- b) Definite volume only
- c) No volume

d) No shape and no volume

Answer: b) Definite volume only

53. Gases have

a) Strong intermolecular force

b) Definite shape

c) Definite volume

d) Very weak intermolecular force

Answer: d) Very weak intermolecular force

54. Intermolecular space is maximum in

a) Solids

b) Liquids

c) Gases

d) Plasmas

Answer: c) Gases

55. On heating, substances generally

a) Contract

b) Expand

c) Freeze

d) Condense

Answer: b) Expand

56. Thermal expansion is minimum in

a) Gases

b) Liquids

c) Solids

d) Vapours

Answer: c) Solids

57. Liquids expand more than solids because

- a) Strong forces
- b) No particles
- c) Less intermolecular force
- d) Fixed shape

Answer: c) Less intermolecular force

58. Gases expand the most because

- a) Particles are tightly packed
- b) Particles move freely
- c) Forces are strong
- d) Volume is fixed

Answer: b) Particles move freely

59. Change of state is a

- a) Chemical change
- b) Irreversible change
- c) Physical change
- d) Nuclear change

Answer: c) Physical change

60. Melting is the change from

- a) Liquid to gas
- b) Solid to liquid
- c) Gas to liquid
- d) Solid to gas

Answer: b) Solid to liquid

61. Freezing is the change from

- a) Gas to liquid

b) Liquid to solid

c) Solid to gas

d) Liquid to gas

Answer: b) Liquid to solid

62. Sublimation is the change from

a) Solid to liquid

b) Liquid to gas

c) Solid to gas

d) Gas to solid

Answer: c) Solid to gas

63. Condensation is the change from

a) Solid to gas

b) Gas to liquid

c) Liquid to solid

d) Solid to liquid

Answer: b) Gas to liquid

64. Vaporization is the change from

a) Solid to gas

b) Liquid to gas

c) Gas to solid

d) Liquid to solid

Answer: b) Liquid to gas

65. Deposition is the change from

a) Liquid to solid

b) Gas to solid

c) Solid to liquid

d) Gas to liquid

Answer: b) Gas to solid

66. Which state has fixed volume but no fixed shape?

a) Solid

b) Liquid

c) Gas

d) Plasma

Answer: b) Liquid

67. Which state has neither fixed shape nor fixed volume?

a) Solid

b) Liquid

c) Gas

d) Ice

Answer: c) Gas

68. Intermolecular force is strongest in

a) Gases

b) Liquids

c) Solids

d) Vapours

Answer: c) Solids

69. Which process does not change chemical composition?

a) Burning

b) Rusting

c) Melting

d) Digestion

Answer: c) Melting

70. Heating a gas mainly increases

- a) Mass
- b) Volume
- c) Colour
- d) Chemical nature

Answer: b) Volume

71. Which of the following has definite shape?

- a) Air
- b) Water
- c) Ice
- d) Steam

Answer: c) Ice

72. Particles in liquids can

- a) Only vibrate
- b) Not move
- c) Slide past one another
- d) Move freely like gas

Answer: c) Slide past one another

73. Expansion on heating is due to

- a) Loss of particles
- b) Increase in particle energy
- c) Chemical reaction
- d) Change in colour

Answer: b) Increase in particle energy

74. Which change is reversible?

- a) Burning paper

- b) Rusting iron
- c) Melting ice
- d) Cooking food

Answer: c) Melting ice

75. Matter can be sensed by

- a) Only eyes
- b) Only touch
- c) Five senses
- d) Sixth sense

Answer: c) Five senses

76. Which is a solid non-metal?

- a) Oxygen
- b) Bromine
- c) Carbon
- d) Neon

Answer: c) Carbon

77. Which metal is used for galvanization?

- a) Copper
- b) Zinc
- c) Aluminium
- d) Iron

Answer: b) Zinc

78. Which element is most abundant in air?

- a) Oxygen
- b) Carbon dioxide
- c) Nitrogen

d) Argon

Answer: c) Nitrogen

79. Which of the following is a compound?

a) Oxygen

b) Nitrogen

c) Water

d) Helium

Answer: c) Water

80. The smallest particle of an element that retains its properties is

a) Molecule

b) Ion

c) Atom

d) Compound

Answer: c) Atom